

XAI Method Comparison (Grad-CAM / Grad-CAM++ / LIME / SHAP)

PAUT Explainable-AI Digital Twin — weld defect inspection from Total Focusing Method (TFM) images.

Generated by `src.xai.compare\_methods` on 6 val images. Every method explains the same scalar (predicted probability of the true defect class over the segmented region) and is scored by the same faithfulness metrics.

Overall trust score (mean across all images)

Method	Trust	Deletion (↓)	Insertion (↑)
Grad-CAM++	0.967	0.048	0.950
Grad-CAM	0.951	0.060	0.912
LIME	0.633	0.049	0.949
SHAP	0.574	0.133	0.854

Trust score by defect class

Method	porosity	slag
Grad-CAM++	0.968	0.967
Grad-CAM	0.951	0.950
LIME	0.632	0.634
SHAP	0.516	0.631

Winner: Grad-CAM++ (trust 0.967). Chart: data/processed/xai/method_comparison.png.

Interpretation

- **The gradient CAMs win because their peak lands on the defect every time** (pointing-game

1.00), while LIME and SHAP score 0.00 on pointing-game on every image — their attribution *peak* falls outside the true defect even when their deletion/insertion looks decent. LIME's superpixels are too coarse for fine TFM speckle and SHAP's blur-masking smears credit off the small defect, so neither localises the defect precisely.

- **LIME/SHAP are not "broken" — they are the wrong tool here.** They were designed for

classification, not a per-pixel segmentation mask; this experiment confirms empirically what the architecture doc argued, instead of just asserting it.

- **Per class:** the ranking holds for both porosity and slag; SHAP is weakest on porosity (0.516) — the smaller, fainter defect, where blur-masking hurts most.

- **Decision:** keep **Grad-CAM / Grad-CAM++ (Seg-Grad-CAM)** on the **segmentation head** as the

project's explainer; LIME/SHAP are reported here as benchmarked alternatives, not adopted.